§ 117. The Convergents

If we transform a continued fraction

$$x = (a_0, a_1, a_2, \ldots, a_{n-1}, x_n), \tag{1}$$

where x_n may be regarded as a variable quantity, into an ordinary fraction, then numerator and denominator become linear expressions in x_n. Thus

$$x = \frac{P_n x_n + P_{n-1}}{Q_n x_n + Q_{n-1}}, \tag{2}$$

where $P_n, Q_n, P_{n-1}, Q_{n-1}$ are integral rational functions of $a_0, a_1, a_2, \ldots, a_{n-1}$. This is verified for the first values $n = 1, 2, \ldots$ by direct calculation:

$$x = a_0 + \frac{1}{x_1} = \frac{a_0 x_1 + 1}{x_1},$$

and

$$x = a_0 + \cfrac{1}{a_1 + \cfrac{1}{x_2}} = \frac{(a_0 a_1 + 1)x_2 + a_0}{a_1 x_2 + 1}.$$

If formula (2) has already been proved for n, then, putting

$$x_n = a_n + \frac{1}{x_{n+1}},$$

one obtains

$$x = \frac{(P_n a_n + P_{n-1})x_{n+1} + P_n}{(Q_n a_n + Q_{n-1})x_{n+1} + Q_n}.$$

This is precisely the formula obtained from (2) by replacing n by $n+1$, provided the following law of formation is adopted:

$$P_{n+1} = a_n P_n + P_{n-1}, \qquad Q_{n+1} = a_n Q_n + Q_{n-1}. \tag{3}$$

These formulae determine P_n, Q_n completely for $n = 2, 3, 4, \ldots$, once one adds

$$P_0 = 1, \quad P_1 = a_0, \qquad Q_0 = 0, \quad Q_1 = 1. \tag{4}$$

The fractions

$$\frac{P_0}{Q_0}, \quad \frac{P_1}{Q_1}, \quad \frac{P_2}{Q_2}, \ldots,$$

of which the first, with denominator zero, is introduced only formally for uniformity, are called the convergents of the continued fraction (1); P_n is the numerator and Q_n the denominator of the n-th convergent.

The formation law (3) shows that if the a's are integers, then P_n and Q_n are integers. If the partial denominators from a_1 onward are positive, then the denominators $Q_1, Q_2, Q_3, \ldots$ are all positive and increase with n; since they are integers, they increase without bound. Only in the special case $a_1 = 1$ does one have $Q_1 = Q_2 = 1$, and the increase then begins only with Q_3. We may therefore state:

II. If Q_{n-1}, Q_n are the denominators of two consecutive convergents, then

$$0 \leq Q_{n-1} \leq Q_n, \qquad Q_n \to \infty \quad (n \to \infty),$$

where equality with the lower bound 0 can occur only for $n = 1$, and equality $Q_{n-1} = Q_n$ can occur only for $n = 2$.

The $P_1, P_2, P_3, \ldots$ are likewise all positive if a_0 is positive; if a_0 is negative, then they are all negative, except possibly $P_2 = a_0 a_1 + 1$, which may be zero. The Q_n are quite independent of a_0. The way in which the P_n depend on a_0 can be seen as follows.

Take the sequence

$$a_1, a_2, a_3, a_4, \ldots$$

as given, and consider the recurrence

$$T_{n+1} = a_n T_n + T_{n-1}. \tag{5}$$

Then T_n is completely determined for all n once T_0 and T_1 are given.

Consider two special cases, Q_n and R_n, determined by

$$R_0 = 1, \quad R_1 = 0, \qquad Q_0 = 0, \quad Q_1 = 1. \tag{6}$$

The general solution of (5) is then

$$T_n = T_0 R_n + T_1 Q_n, \tag{7}$$

and this includes in particular

$$P_n = R_n + a_0 Q_n. \tag{8}$$

If T_n and S_n are two solutions of (5), so that

$$T_{n+1} = a_n T_n + T_{n-1}, \qquad S_{n+1} = a_n S_n + S_{n-1},$$

then elimination of a_n gives

$$T_{n+1} S_n - T_n S_{n+1} = -(T_n S_{n-1} - T_{n-1} S_n).$$

Thus, putting

$$T_n S_{n+1} - S_n T_{n+1} = \Delta_n,$$

one has

$$\Delta_n = -\Delta_{n-1} = \Delta_{n-2} = \cdots = (-1)^n \Delta_0. \tag{9}$$

For $T_n = R_n$ and $S_n = Q_n$ one has, by (6), $\Delta_0 = 1$; hence

$$R_n Q_{n+1} - Q_n R_{n+1} = (-1)^n.$$

Replacing n by $n-1$ and changing the sign, one obtains

$$R_n Q_{n-1} - Q_n R_{n-1} = (-1)^n. \tag{10}$$

Applying this to two functions T_n, S_n of the form

$$T_n = T_0 R_n + T_1 Q_n, \qquad S_n = S_0 R_n + S_1 Q_n,$$

one obtains

$$T_n S_{n-1} - S_n T_{n-1} = (-1)^n (T_0 S_1 - T_1 S_0),$$

and in particular

$$P_n Q_{n-1} - Q_n P_{n-1} = (-1)^n. \tag{11}$$

This formula, which will be used repeatedly, shows that P_n and Q_n have no common divisor; thus the convergent $P_n : Q_n$ cannot be reduced by cancellation.

If the continued fraction is finite, that is, if the final quantity $x_{n-1} = a_{n-1}$ is an integer, then the formation of convergents is not continued beyond

$$\frac{P_n}{Q_n} = \frac{P_{n-1} a_{n-1} + P_{n-2}}{Q_{n-1} a_{n-1} + Q_{n-2}}.$$

Formula (2) then shows that the last convergent agrees with the value of the continued fraction.

§ 118. Solution of Indeterminate Equations with Two Unknowns

The formulae just obtained lead to the solution of a problem that occurs in many applications:

> Let α, β be two given integers without a common divisor. We seek two other integers satisfying
> $$\alpha y - \beta x = 1. \tag{1}$$

If this problem is soluble at all, it has infinitely many solutions, all derived from any one of them.

Indeed, if x_0, y_0 is a solution, so that

$$\alpha y_0 - \beta x_0 = 1, \tag{2}$$

then subtracting (2) from (1) gives

$$\alpha(y - y_0) = \beta(x - x_0).$$

Since α has no common divisor with β, it follows that $x - x_0$ is divisible by α. Thus, writing h for an arbitrary integer, put

$$x = x_0 + h\alpha;$$

then

$$y = y_0 + h\beta,$$

and all solutions of (1) are contained in this form.

It remains only to find one solution of (1). We can always do this as follows. We suppose β positive, which is no essential restriction, since if necessary x may be replaced by $-x$, and expand the rational number $\alpha : \beta$ into a continued fraction according to §. 115:

$$\frac{\alpha}{\beta} = (a_0, a_1, a_2, \ldots, a_{n-1}).$$

Let its convergents be

$$\frac{P_0}{Q_0}, \quad \frac{P_1}{Q_1}, \quad \frac{P_2}{Q_2}, \ldots, \frac{P_{n-1}}{Q_{n-1}}, \quad \frac{P_n}{Q_n},$$

so that

$$\frac{\alpha}{\beta} = \frac{P_n}{Q_n}.$$

Since α, β and also P_n, Q_n are relatively prime, and since β and Q_n are positive, it follows that

$$\alpha = P_n, \qquad \beta = Q_n.$$

Furthermore, by §. 117,

$$P_n Q_{n-1} - Q_n P_{n-1} = (-1)^n,$$

and hence

$$x = (-1)^n P_{n-1}, \qquad y = (-1)^n Q_{n-1} \tag{3}$$

are integers satisfying (1). The problem is therefore completely solved.

For moderately large numbers the calculation is quite simple. For example, set $\alpha = 24335$, $\beta = 3588$. One first obtains, by §. 115,

$$\frac{24335}{3588} = (6, 1, 3, 1, 1, 2, 6, 2, 1, 1, 4),$$

and the convergents

$$\frac{1}{0}, \quad \frac{6}{1}, \quad \frac{7}{1}, \quad \frac{27}{4}, \quad \frac{34}{5}, \quad \frac{61}{9}, \quad \frac{156}{23}, \quad \frac{997}{147}, \quad \frac{2150}{317}, \quad \frac{3147}{464}, \quad \frac{5297}{781}, \quad \frac{24335}{3588}.$$

Here $n = 11$ is odd, and therefore one must put

$$x = -5297, \qquad y = -781.$$

These are the absolutely smallest values of x, y satisfying (1).

The least positive solution is obtained from them by adding α and β respectively:

$$x = 19038, \qquad y = 2807.$$

The solution in integers x, y of the more general equation

$$\alpha y - \beta x = \gamma \tag{4}$$

is reduced to this case.

First, it is clear that, if (4) is to be soluble, every common divisor of α and β must also divide γ. Such a common divisor is then removed by division. We therefore again assume that α and β have no common divisor. Then, exactly as above, all solutions of (4) are obtained from one of them, x_0, y_0, in the form

$$x = x_0 + h\alpha, \qquad y = y_0 + h\beta,$$

where h is an arbitrary integer. If in (4) one writes

$$x = \gamma\xi, \qquad y = \gamma\eta,$$

and divides by γ, then (4) becomes

$$\alpha\eta - \beta\xi = 1,$$

i.e. an equation of the form (1).

The solution becomes completely determined if an additional condition is given from which h can be determined, for example that y should lie between 0 and β, including one of the endpoints. We summarize this in the following theorem:

III. The indeterminate equation

$$\alpha y - \beta x = \gamma$$

always has an integral solution x, y when α, β, γ are integers and α and β have no common divisor; and it has only one such solution if, in addition, one imposes either

$$0 \leq y < \beta,$$

or

$$0 < y \leq \beta.$$

From the preceding remarks, equation (4) also always has solutions when γ is divisible by the greatest common divisor of α and β. In particular:

The greatest common divisor δ of two numbers α, β can be represented in the form

$$\alpha x - \beta y = \delta,$$

where x, y are integers.

Theorem III may be generalized as follows:

If $\alpha_1, \alpha_2, \alpha_3, \ldots$ are any finite number of given integers without a divisor common to all, and m is any given integer, then integers $x_1, x_2, x_3, \ldots$ can be determined so that

$$m = \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \cdots . \tag{5}$$

If the right hand side of (5) consists of only two terms, this is just Theorem III. Suppose, then, that the possibility of satisfying (5) has already been proved when the right hand side contains fewer terms. If δ is the greatest common divisor of $\alpha_2, \alpha_3, \ldots$, then δ must be relatively prime to α_1; hence the two equations

$$\delta = \alpha_2 y_2 + \alpha_3 y_3 + \cdots, \qquad m = \alpha_1 x + \delta x_1$$

can be satisfied by integers $x, x_1, y_2, y_3, \ldots$. Then (5) is satisfied by

$$x_1 = x, \qquad x_2 = x_1 y_2, \qquad x_3 = x_1 y_3, \ldots.$$

In number theory, the problem solved by Theorem III is also expressed as follows: given α, β, γ, find a number y satisfying the congruence

$$\alpha y \equiv \gamma \pmod{\beta}.$$

This problem therefore always has a solution when γ is divisible by the greatest common divisor of α and β.

§ 119. Convergence of the Convergents

We now assume that the sequence $a_1, a_2, a_3, \ldots$ is unbounded in length, and form the difference of two consecutive convergents:

$$\frac{P_n}{Q_n} - \frac{P_{n-1}}{Q_{n-1}} = \frac{(-1)^n}{Q_n Q_{n-1}}. \tag{1}$$

This difference is positive for even n, negative for odd n, and decreases indefinitely in absolute value as n increases without bound. From (1) we further obtain

$$\begin{aligned} \frac{P_n}{Q_n} - \frac{P_{n-2}}{Q_{n-2}} &= \frac{(-1)^n}{Q_n Q_{n-1}} + \frac{(-1)^{n+1}}{Q_{n-1} Q_{n-2}} \\ &= \frac{(-1)^n}{Q_{n-1}} \left(\frac{1}{Q_n} - \frac{1}{Q_{n-2}} \right). \end{aligned} \tag{2}$$

Since $Q_{n-2} < Q_n$, this difference is negative for even n and positive for odd n.

It follows that the sequence of convergents with even index

$$\frac{P_2}{Q_2}, \quad \frac{P_4}{Q_4}, \quad \frac{P_6}{Q_6}, \ldots \tag{3}$$

is decreasing, while the sequence of convergents with odd index

$$\frac{P_1}{Q_1}, \quad \frac{P_3}{Q_3}, \quad \frac{P_5}{Q_5}, \ldots \tag{4}$$

is increasing.

By §. 117,

$$x = \frac{P_n x_n + P_{n-1}}{Q_n x_n + Q_{n-1}},$$

and therefore

$$\frac{P_n}{Q_n} - x = \frac{(-1)^n}{Q_n (Q_n x_n + Q_{n-1})}. \tag{5}$$

Since Q_n, Q_{n-1}, x_n are positive, all numbers in the sequence (3) are greater than x, and all numbers in the sequence (4) are smaller than x.

The difference (5) tends below every bound as n increases without limit. Since $x_n > a_n$, and hence $Q_n x_n + Q_{n-1} > Q_{n+1}$, its absolute value is smaller than

$$\frac{1}{Q_n Q_{n+1}}, \tag{6}$$

and a fortiori smaller than $1 : Q_n^2$.

Thus the numbers in the sequence (3) approach the limit x from above, and those in (4) from below. Expression (6) gives a measure of the error made when x is replaced by the convergent $P_n : Q_n$.

> The convergents are therefore approximate expressions of irrational numbers by rational fractions.

The fact that, for the same degree of approximation to the irrational number x, these convergents are the simplest possible is expressed by the following theorem:

> IV. Between the two rational fractions
>
> $$\frac{P_n}{Q_n}, \qquad \frac{P_{n-1}}{Q_{n-1}}$$
>
> there cannot be inserted any other rational fraction whose denominator is less than Q_n, or even equal to Q_n.

Suppose that the rational fraction $M : N$ lies between the convergents $P_n : Q_n$ and $P_{n-1} : Q_{n-1}$. Then the difference

$$\frac{P_n}{Q_n} - \frac{P_{n-1}}{Q_{n-1}}$$

is greater in absolute value and has the same sign as

$$\frac{M}{N} - \frac{P_{n-1}}{Q_{n-1}}.$$

Hence

$$(-1)^n \left(\frac{P_n}{Q_n} - \frac{P_{n-1}}{Q_{n-1}} \right) > (-1)^n \left(\frac{M}{N} - \frac{P_{n-1}}{Q_{n-1}} \right).$$

Multiplying both sides by NQ_{n-1} and using (1), we get

$$\frac{N}{Q_n} > (-1)^n (MQ_{n-1} - NP_{n-1}).$$

The right hand side is a positive integer, hence at least 1, and therefore

$$N > Q_n.$$

§ 120. Equivalent Numbers

Continued fractions lead us to consider a special kind of linear substitution, one that is of great importance in algebra and number theory generally.

Let x and y be two numbers, or variable quantities, related by

$$y = \frac{\alpha x + \beta}{\gamma x + \delta}, \tag{1}$$

where $\alpha, \beta, \gamma, \delta$ are integers satisfying

$$\alpha\delta - \beta\gamma = \varepsilon = \pm 1. \tag{2}$$

Then we call x and y equivalent. We call them properly or improperly equivalent according as $\varepsilon = +1$ or $\varepsilon = -1$. This relation is reciprocal, for (1) gives

$$x = \frac{\delta y - \beta}{-\gamma y + \alpha}. \tag{3}$$

If two quantities are equivalent to a third, then they are equivalent to each other. For if

$$x = \frac{\alpha' z + \beta'}{\gamma' z + \delta'}, \tag{4}$$

then (1) gives

$$y = \frac{\alpha'' z + \beta''}{\gamma'' z + \delta''}, \tag{5}$$

where

$$\begin{aligned} \alpha'' &= \alpha\alpha' + \beta\gamma', \quad & \beta'' &= \alpha\beta' + \beta\delta', \\ \gamma'' &= \gamma\alpha' + \delta\gamma', \quad & \delta'' &= \gamma\beta' + \delta\delta', \end{aligned} \tag{6}$$

and

$$\alpha''\delta'' - \beta''\gamma'' = (\alpha\delta - \beta\gamma)(\alpha'\delta' - \beta'\gamma'), \tag{7}$$

or

$$\varepsilon'' = \varepsilon\varepsilon'. \tag{8}$$

Substitution (5) is called the composition of (1) and (4).

There is a need for an abbreviated notation for these substitutions. Since often it is not the variables but only the substitution numbers $\alpha, \beta, \gamma, \delta$ that matter, the whole substitution (1) is denoted by its substitution numbers, or simply by a letter:

$$S = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \tag{9}$$

When needed, equation (1) is then written

$$y = S(x) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (x).$$

The composition of two substitutions is denoted by juxtaposition of the symbols, but attention must be paid to the order:

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} = \begin{pmatrix} \alpha'' & \beta'' \\ \gamma'' & \delta'' \end{pmatrix}, \tag{10}$$

or

$$SS' = S''. \tag{11}$$

Formulae (6) give the rule for forming the composite substitution. They can be derived from the multiplication rule for determinants, but one must observe that in the first factor one sums by rows and in the second by columns, as the formulae (6) show.

Just as two substitutions can be composed, one may also compose several:

$$SS_1S_2S_3 \cdots .$$

In forming the composite substitution, the components cannot in general be interchanged; but one may combine any two neighboring components into one, then again combine two, and so on. Thus the commutative law does not hold, but the associative law does.

This follows immediately from the fact that, if x is expressed through x_1, x_1 through x_2, and x_2 through x_3, then the expression of x through x_3 may be found either by first substituting the expression of x_1 in terms of x_2 into x, and then substituting x_2 in terms of x_3, or by first expressing x_1 through x_3 and then substituting this in the expression of x through x_1.

By (8), a composite equivalence is proper or improper according as an even or odd number of its components are improper.

If the two substitutions

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad \begin{pmatrix} \varepsilon\delta & -\varepsilon\beta \\ -\varepsilon\gamma & \varepsilon\alpha \end{pmatrix} \tag{12}$$

are composed, one obtains, regardless of which is placed first,

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{13}$$

By (1), this substitution is equivalent to $y = x$; it changes nothing and is called the identical substitution, also denoted briefly by (1). Thus the two substitutions (12) are called reciprocal to one another and are denoted

$$S, \quad S^{-1},$$

or one writes

$$\begin{pmatrix} \varepsilon\delta & -\varepsilon\beta \\ -\varepsilon\gamma & \varepsilon\alpha \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}^{-1}. \tag{14}$$

Composition with the identical substitution (13) leaves every other substitution unchanged.

From this we very simply derive the proof of the theorem:

V. All rational numbers are equivalent to one another, both properly and improperly.

Let $m : n$ and $m' : n'$ be two rational fractions, with m, n and m', n' respectively relatively prime. If $\varepsilon, \varepsilon'$ are chosen arbitrarily as ± 1, then by §. 118 the numbers μ, ν and μ', ν' may be chosen so that

$$m\nu - n\mu = \varepsilon, \qquad m'\nu' - n'\mu' = \varepsilon'.$$

Thus

$$\begin{pmatrix} m & \mu \\ n & \nu \end{pmatrix}, \qquad \begin{pmatrix} m' & \mu' \\ n' & \nu' \end{pmatrix}$$

are two linear substitutions. Hence

$$\begin{pmatrix} m & \mu \\ n & \nu \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} m' & \mu' \\ n' & \nu' \end{pmatrix}$$

is also a linear substitution, and

$$m = \alpha m' + \beta n', \qquad n = \gamma m' + \delta n'.$$

Therefore

$$\frac{m}{n} = \frac{\alpha \frac{m'}{n'} + \beta}{\gamma \frac{m'}{n'} + \delta},$$

as required. The equivalence is proper or improper according as $\varepsilon = \varepsilon'$ or $\varepsilon = -\varepsilon'$.

Every number is improperly equivalent to its negative and to its reciprocal, since

$$-x = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}(x), \qquad \frac{1}{x} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}(x).$$

§ 121. Expansion of Equivalent Numbers in Continued Fractions

If, of two equivalent numbers, one is irrational, then the other is also irrational; and if one is real, so is the other. We make these two assumptions and therefore put, in

$$y = \frac{\alpha x + \beta}{\gamma x + \delta}, \qquad \alpha\delta - \beta\gamma = \varepsilon, \tag{1}$$

x as irrational and real. We now investigate how the continued fraction expansion of y is derived from that of x.

Let x be developed into a continued fraction with final quantity x_n:

$$x = (a_0, a_1, a_2, \ldots, a_{n-1}, x_n), \tag{2}$$

where for the moment n remains undetermined.

Expressed by the convergents,

$$x = \frac{P_n x_n + P_{n-1}}{Q_n x_n + Q_{n-1}}, \qquad P_n Q_{n-1} - Q_n P_{n-1} = (-1)^n. \tag{3}$$

Substitution in (1) gives

$$y = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} P_n & P_{n-1} \\ Q_n & Q_{n-1} \end{pmatrix} (x_n) = \begin{pmatrix} R_n & R_{n-1} \\ S_n & S_{n-1} \end{pmatrix} (x_n), \tag{4}$$

where, by §. 120, (10) and (6), (7),

$$\begin{aligned} R_n &= \alpha P_n + \beta Q_n, \quad R_{n-1} = \alpha P_{n-1} + \beta Q_{n-1}, \\ S_n &= \gamma P_n + \delta Q_n, \quad S_{n-1} = \gamma P_{n-1} + \delta Q_{n-1}, \end{aligned} \tag{5}$$

$$R_n S_{n-1} - S_n R_{n-1} = (-1)^n \varepsilon. \tag{6}$$

Put S_n in the form

$$S_n = Q_n \left(\gamma \frac{P_n}{Q_n} + \delta \right).$$

Since $P_n : Q_n$ approaches x to arbitrary accuracy and Q_n is positive, S_n has, for sufficiently large n, the same sign as

$$\gamma x + \delta.$$

As x is irrational, this quantity is not zero, and we shall suppose it positive. If it were negative, one would only have to change the signs of all four numbers $\alpha, \beta, \gamma, \delta$ simultaneously, which leaves (1) unchanged, in order to make it positive.

Thus S_n is positive for sufficiently large n. From (5), with §. 117, one obtains

$$S_{n+1} = a_n S_n + S_{n-1};$$

it follows that if n is so large that S_{n-2} and the following S's are positive, then S_n increases with n, hence

$$S_n > S_{n-1} > 0. \tag{7}$$

Now expand the rational fraction $R_n : S_n$ into a finite continued fraction

$$\frac{R_n}{S_n} = (b_0, b_1, b_2, \ldots, b_{m-1}), \tag{8}$$

and denote the penultimate convergent by $R' : S'$, so that

$$R_n S' - S_n R' = (-1)^m. \tag{9}$$

By Proposition I of §. 115, however, m may be chosen even or odd at will, and we choose it so that

$$(-1)^m = (-1)^n \varepsilon.$$

Moreover, by §. 117, II,

$$S_n \geq S' \geq 0, \tag{10}$$

where equality at the lower boundary can occur only for $m = 1$, and equality at the upper boundary only for $m = 2$ and $b_1 = 1$, that is, only when $S_n = 1$. This can be avoided by taking n sufficiently large.

By §. 118, III the numbers S', R' are completely determined by (9), (10); and since S_{n-1}, R_{n-1} satisfy the same conditions by (6), (7), it follows that

$$S' = S_{n-1}, \qquad R' = R_{n-1}.$$

Consequently the continued fraction with final quantity x_n

$$(b_0, b_1, b_2, \ldots, b_{m-1}, x_n) \tag{11}$$

has the value

$$\frac{R_n x_n + R_{n-1}}{S_n x_n + S_{n-1}},$$

and therefore agrees with y.

If x_n is now further expanded into a continued fraction

$$x_n = (a_n, a_{n+1}, a_{n+2}, \ldots),$$

then (2) and (11) give

$$x = (a_0, a_1, a_2, \ldots, a_{n-1}, a_n, a_{n+1}, a_{n+2}, \ldots),$$
$$y = (b_0, b_1, b_2, \ldots, b_{m-1}, a_n, a_{n+1}, a_{n+2}, \ldots).$$

In words:

> VI. The continued fraction expansions of two equivalent numbers agree from some partial denominator onward.

We may add that if the equivalence is proper ($\varepsilon = +1$), then the number of non-agreeing preceding partial denominators is even in both expansions or odd in both; whereas if the equivalence is improper, this number is even in one expansion and odd in the other.

The converse is easy to see. If two continued fractions agree from a certain partial denominator onward, then they can be written so as to have the same final quantity. But the value of a continued fraction is always equivalent to each of its final quantities [§. 117, (2)]; hence two continued fractions with the same final quantity are equivalent to one another.

§ 122. Quadratic Irrational Numbers

Simple and beautiful laws result when the continued fraction expansion is applied to the determination of the roots of a quadratic equation. The root of an integral quadratic equation has the form

$$\omega = x + y\sqrt{d}, \tag{1}$$

where x, y, d are integral or fractional rational numbers. Without restricting the generality, however, d may be assumed to be an integer without a square divisor; for if d has a denominator, one can clear it and count the root of the square denominator, and any square divisor of the numerator, into y. We further assume that $d \neq 1$, for otherwise ω would be rational. The number d may be positive or negative, and this determines whether ω is real or imaginary.

Changing the sign of the root gives

$$\omega' = x - y\sqrt{d}, \tag{2}$$

which is called the conjugate number to ω. The product

$$\omega\omega' = x^2 - y^2 d$$

of the two numbers ω, ω' is called the norm of ω (or also of ω'). The norm of a product of two quadratic irrational numbers is equal to the product of their norms.

The numbers ω and ω' are the roots of an equation with rational coefficients:

$$\omega^2 - 2x\omega + (x^2 - y^2 d) = 0. \tag{3}$$

To transform this into an integral equation, put

$$x^2 - y^2 d = -\frac{a}{c}, \qquad 2x = \frac{b}{c}, \tag{4}$$

where a, b, c are integers without a common divisor. Then (3) becomes

$$c\omega^2 = a + b\omega, \tag{5}$$

a primitive integral equation.

The numbers a, b, c are determined by (4) only up to a common sign. The discriminant of (3) is

$$D = b^2 + 4ac = 4c^2y^2d, \tag{6}$$

and it is also to be called the discriminant of the irrational number ω. Thus D is an integer with the same sign as d, and it is divisible by d. Indeed, if p is a prime divisor of d, which by hypothesis occurs in d only to the first power, then p cannot occur in the denominator of $2cy$ and must therefore occur in D by (6). Hence $2cy$ is an integer and the quotient $D : d$ is a square.

From (4) and (6), ω and ω' may be written

$$\omega = \frac{\sqrt{D} + b}{2c}, \qquad \omega' = \frac{-\sqrt{D} + b}{2c}, \tag{7}$$

or equivalently

$$\omega = \frac{2a}{\sqrt{D} - b}, \qquad \omega' = \frac{-2a}{\sqrt{D} + b}, \tag{8}$$

where the sign of the root is fixed by

$$\sqrt{D} = 2cy\sqrt{d}. \tag{9}$$

Since an even square is divisible by 4, while an odd square leaves remainder 1 upon division by 4, it follows from (6) that

$$D \equiv 0 \quad \text{or} \quad D \equiv 1 \pmod 4. \tag{10}$$

We next examine how the numbers a, b, c, D change when one passes from ω to an equivalent number ω_1. Let

$$\omega_1 = \frac{\alpha\omega + \beta}{\gamma\omega + \delta}, \qquad \omega = \frac{\delta\omega_1 - \beta}{-\gamma\omega_1 + \alpha}, \qquad \alpha\delta - \beta\gamma = \varepsilon. \tag{11}$$

From (5), the quadratic equation for ω_1 is

$$c_1\omega_1^2 = a_1 + b_1\omega_1, \tag{12}$$

if

$$\begin{aligned} -a_1 &= -a\alpha^2 + b\alpha\beta + c\beta^2, \\ b_1 &= -2a\alpha\gamma + b(\alpha\delta + \beta\gamma) + 2c\beta\delta, \\ c_1 &= -a\gamma^2 + b\gamma\delta + c\delta^2. \end{aligned} \tag{13}$$

Solving (13) for a, b, c gives

$$\begin{aligned} a &= a_1\delta^2 + b_1\beta\delta - c_1\beta^2, \\ b &= 2a_1\gamma\delta + b_1(\alpha\delta + \beta\gamma) - 2c_1\alpha\beta, \\ -c &= -a_1\gamma^2 + b_1\alpha\gamma - c_1\alpha^2, \end{aligned} \tag{14}$$

and also

$$b^2 + 4ac = b_1^2 + 4a_1c_1 = D. \tag{15}$$

Since a, b, c have no common divisor, neither do a_1, b_1, c_1, as is seen from (14); and a_1, b_1, c_1 have the same meaning for ω_1 that a, b, c have for ω.

> VII. It is further seen from (15) that equivalent numbers have not only the same irrationality $\sqrt{d}$, but also the same discriminant.

Putting

$$\omega_1 = x_1 + y_1\sqrt{d} = \frac{\alpha(x + y\sqrt{d}) + \beta}{\gamma(x + y\sqrt{d}) + \delta}, \tag{16}$$

and multiplying the last fraction by $\gamma(x - y\sqrt{d}) + \delta$ in order to determine x_1, y_1, one obtains from (4) and (13)

$$2x_1 = \frac{b_1}{c_1}, \qquad y_1 = \frac{\varepsilon y c}{c_1}. \tag{17}$$

Thus, if $\sqrt{D}$ is understood as in (7) and (8), then by (9)

$$\omega_1 = \frac{b_1 + \varepsilon\sqrt{D}}{2c_1}. \tag{18}$$

§ 123. Reduced Numbers with Negative Discriminant

The preceding considerations apply equally to the two cases of real and imaginary quadratic irrational numbers, that is, to positive and negative discriminants. Now, however, the two cases must be separated, and we begin with negative discriminants. Thus we are concerned with imaginary numbers

$$\omega = x + y\sqrt{d} = \xi + i\eta, \tag{1}$$

where ξ and η are real. Of the two conjugates $\xi + i\eta$, $\xi - i\eta$, it suffices to consider one. We therefore assume η positive and make the following definition.

A complex number $\omega = \xi + i\eta$ with positive η is called reduced if

$$-\frac{1}{2} \leq \xi \leq \frac{1}{2}, \qquad \xi^2 + \eta^2 \geq 1. \tag{2}$$

If ξ, η are regarded as rectangular coordinates in a plane, then every number ω is represented by a point in this plane. The positions of the points corresponding to reduced numbers are represented by the shaded region in Fig. 26, including the boundary; this region will be called the fundamental region.

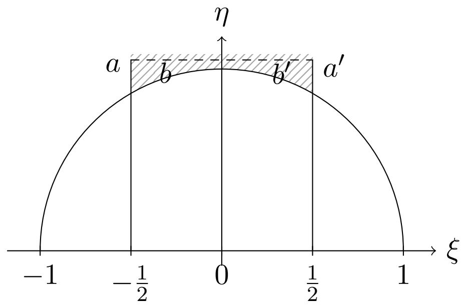

Fig. 26.

The conditions (2) further imply, as is also easily confirmed from the figure,

$$\eta^2 \geq \frac{3}{4}. \tag{3}$$

If one passes from a number ω to an equivalent number ω_1, then formula (16) of the preceding paragraph gives

$$\omega_1 = \frac{\alpha\xi + \beta + i\alpha\eta}{\gamma\xi + \delta + i\gamma\eta} = \frac{[(\alpha\xi + \beta)(\gamma\xi + \delta) + \alpha\gamma\eta^2] + i\varepsilon\eta}{(\gamma\xi + \delta)^2 + \gamma^2\eta^2}. \tag{4}$$

This formula shows that η_1 has the same sign as η, or the opposite sign, according as $\varepsilon = +1$ or $\varepsilon = -1$, that is, according as the equivalence is proper or improper.

If we restrict ourselves to numbers with positive imaginary part, only proper equivalence is relevant. We now prove the fundamental theorem:

1. Every imaginary quadratic irrational number $x + y\sqrt{d}$ with positive imaginary part is equivalent to a reduced number.

Let α be the integer nearest to ξ. Then $\omega - \alpha$ satisfies the first of the conditions (2), namely

$$-\frac{1}{2} \leq \xi - \alpha \leq \frac{1}{2}.$$

If

$$r^2 = (\xi - \alpha)^2 + \eta^2 = (\omega - \alpha)(\omega' - \alpha) \tag{5}$$

is greater than or equal to 1, then $\omega - \alpha$, which is equivalent to ω, is already reduced. Otherwise we apply the continued fraction process and put

$$\omega = \alpha - \frac{1}{\omega_1}, \tag{6}$$

so that $\omega_1 = \xi_1 + i\eta_1$ is also equivalent to ω. We treat ω_1 in the same way; if $\omega_1 - \alpha_1$ is not yet reduced, put

$$\omega_1 = \alpha_1 - \frac{1}{\omega_2}, \tag{7}$$

and so on. Consider the sequence of quantities, formed analogously to (5),

$$r_1^2 = (\xi_1 - \alpha_1)^2 + \eta_1^2, \quad r_2^2 = (\xi_2 - \alpha_2)^2 + \eta_2^2, \ldots.$$

It remains only to show that after finitely many such steps one reaches an r_ν^2 that is equal to or greater than 1.

By (6),

$$r^2 = (\omega - \alpha)(\omega' - \alpha) = \frac{1}{\xi_1^2 + \eta_1^2}.$$

Comparison of imaginary parts on the two sides of (6) therefore gives

$$\eta = r^2\eta_1.$$

Similarly one obtains the chain

$$\eta = r^2\eta_1, \qquad \eta_1 = r_1^2\eta_2, \qquad \eta_2 = r_2^2\eta_3, \ldots. \tag{8}$$

But $i\eta = y\sqrt{d}$; hence, by formula (6) of §. 122,

$$i\eta = \frac{\sqrt{D}}{2c},$$

where c is an integer. Since equivalent numbers have the same discriminant, one likewise has

$$i\eta_1 = \frac{\sqrt{D}}{2c_1}, \qquad i\eta_2 = \frac{\sqrt{D}}{2c_2}, \ldots,$$

where $c, c_1, c_2, \ldots$ is a sequence of positive integers. Thus (8) gives

$$c_1 = r^2 c, \qquad c_2 = r_1^2 c_1, \qquad c_3 = r_2^2 c_2, \ldots. \tag{9}$$

As long as the numbers $r^2, r_1^2, \ldots, r_\nu^2$ are all less than 1, it follows that

$$c > c_1 > c_2 > \cdots > c_{\nu+1}. \tag{10}$$

Since there are only finitely many positive integers smaller than c, this chain must terminate. This proves theorem 1.

The reduced numbers whose points lie on the boundary of the fundamental region are equivalent in pairs, namely

$$\text{a)} \quad \omega = -\frac{1}{2} + i\eta, \qquad \omega_1 = \omega + 1 = \frac{1}{2} + i\eta,$$

(the points a and a' in the figure), and

$$\text{b)} \quad \omega = -\xi + i\eta, \qquad \omega_1 = -\frac{1}{\omega} = \xi + i\eta,$$

when $\xi^2 + \eta^2 = 1$ (the points b and b' in the figure). In addition, the following theorem holds:

2. Apart from the cases a), b), no two reduced numbers are equivalent.

Suppose two non-identical reduced numbers are given,

$$\omega = \xi + \eta i, \qquad \omega_1 = \xi_1 + \eta_1 i, \tag{11}$$

and assume, without loss of generality, that

$$\eta_1 \geq \eta. \tag{12}$$

Then (4) gives

$$\eta_1 = \frac{\eta}{(\gamma\xi + \delta)^2 + \gamma^2\eta^2}, \tag{13}$$

and therefore, by (12),

$$(\gamma\xi + \delta)^2 + \gamma^2\eta^2 \leq 1. \tag{14}$$

Using (3), we also have

$$1 \geq \gamma^2\eta^2 \geq \frac{3}{4}\gamma^2. \tag{15}$$

Thus there are only two possibilities:

$$\text{a)}\quad \gamma = 0, \qquad \text{b)}\quad \gamma = \pm 1.$$

In case a), $\alpha\delta = 1$, so $\alpha = \delta = \pm 1$. Equations (12) and (13) give $\eta_1 = \eta$, and (4) further gives

$$\xi_1 = \xi \pm \beta.$$

This is compatible with the conditions (2) only if either $\xi_1 = \xi$, $\beta = 0$, and hence ω_1 is identical with ω, or else $\beta = \pm 1$ and $\xi_1 = -\xi = \pm\frac{1}{2}$, which is the exceptional case a).

In case b), (14) implies

$$(\xi + \delta)^2 + \eta^2 \leq 1.$$

Thus either $\alpha = 0$, $\delta = 0$, $\beta\gamma = -1$, $\xi^2 + \eta^2 = 1$, $\eta_1 = \eta$, and by (4) $\xi_1 = \pm\alpha - \xi$; since $\xi_1 = \xi$ is excluded, this gives $\xi_1 = -\xi$, the exceptional case b). Or else $\delta = \pm 1$, so $(\xi \pm 1)^2 + \eta^2 \leq 1$; by (3), $(\xi \pm 1)^2 \leq \frac{1}{4}$. But by (2), $|\xi \pm 1|$ cannot be less than $\frac{1}{2}$, and hence $\xi = \mp\frac{1}{2}$, $\eta^2 = \frac{3}{4}$. Consequently, by (13), $\eta_1^2 = \eta^2 = \frac{3}{4}$, and by (2), $\xi_1^2 = \frac{1}{4}$, $\xi_1 = \mp\frac{1}{2}$, a case included under both a) and b).

This proves theorem 2. We add a third theorem:

> 3. For a given negative discriminant D there are only finitely many reduced irrational numbers.

From the representation of irrational numbers in §. 122, (7),

$$\omega = \frac{\sqrt{D} + b}{2c}, \qquad \xi = \frac{b}{2c}, \qquad \eta = \frac{\sqrt{-D}}{2c},$$

and from the inequality (3), $\eta^2 \geq \frac{3}{4}$, it follows that

$$\frac{-D}{4c^2} \geq \frac{3}{4},$$

or

$$c \leq \sqrt{\frac{-D}{3}}.$$

Thus, for given D, there are only finitely many possible values of the positive integer c. Moreover, from

$$-\frac{1}{2} \leq \xi \leq \frac{1}{2}$$

one obtains

$$-c \leq b \leq c.$$

Thus for each value of c there are only finitely many possible values of b, as was required.

Combining all mutually equivalent numbers into a class, theorem 1 therefore gives:

> 4. The number of classes of imaginary quadratic irrational numbers of a given discriminant is finite.